TLAB TECHNICAL WHITEPAPER · 2026 EDITION

Breaking the 4°C & Acid Barrier

The Next Breakthrough in Premium RTD & Desserts

25%
Real Juice Content

10mm
Industry-LargestDia.

6M
@4°C Shelf Life

393
Custom Flavours

MODULE 1 · MARKET INSIGHTS

Why Traditional Toppings

Fail Premium Markets

$1.06B
Global Market Size by 2032

10.7%
CAGR 2024–2032

29.9%
High-Margin Blue-Ocean Share

63%
Shoppers Prefer Clean-Label / Natural Ingredients

High-margin applications (Frozen Desserts, Yogurt, RTD, Fine Dining) CAGR > 12%

Cold Chain Failure

Starch pearls harden & age at 4°C refrigeration

Acid Degradation

Membrane ruptures in low-pH yogurt / juice

Alcohol Dissolution

Cannot maintain spherical structure in cocktails

Compliance Barrier

Fails allergen-free & clean-label EU/US standards

Reverse Spherification

Molecular Micro-encapsulation Engineering

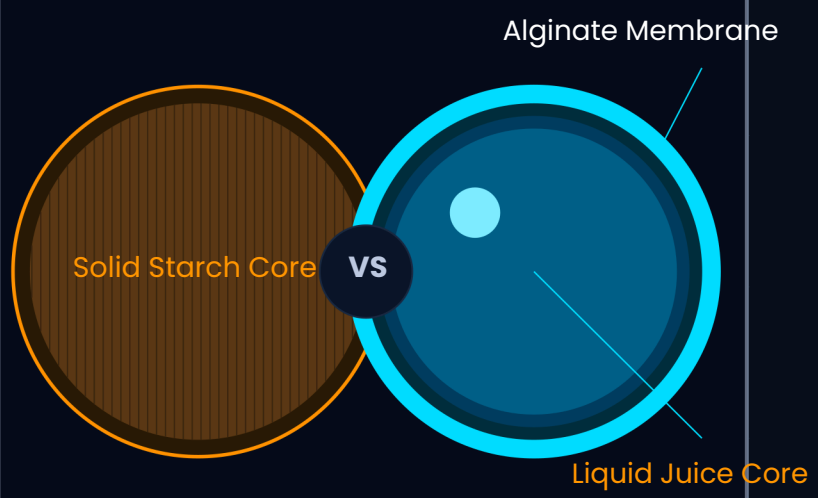
Tachiz abandons traditional starch-cooking, employing Spanish molecular gastro nomy. Sodium alginate cross-links precisely with calcium ions to form a high-resil ience micro-encapsulation membrane — delivering a signature bursting sensatio n, and fully compatible with RTD secondary processing lines.

25%
Real Juice
Industry-highest juice ratio

10mm
Diameter
Industry-largest sphere size

RTD
Ready
Compatible with secondary processing

Structure Cross-Section /



Traditional Pearl **Tlab Popping Boba**

Secondary Processing Compatible

Perfect Survivor in Extreme Environments

ShelfLife&StabilityTest—TlabInternalQCReport2024–2025

6 Mo	25 %	10 mm	120 °C
Cold Shelf Life	Real Juice	Largest Dia.	Heat Tolerance

Cold Resistance — 6 Months

4°C refrigeration · 6 months, no discolouration, stays crunchy

Acid Stability — pH 3.5

Yogurt / juice · Membrane integrity fully maintained

Alcohol Compat. — ABV

Fine cocktails · 3D spherical shape fully maintained

Heat Tolerance — 120°C

120°C commercial sterilisation · RTD line compatible

* Source: Tachiz Tlab internal QC test reports 2024–2025 · Tests conducted under standardised closed-system conditions.

MODULE 4 · FLAVOUR MATRIX & CUSTOMIZATION

Bite Into Extra ___

393 Formulas × 12 Package Formats · Imagination is the only limit

393+
Flavours
Custom flavour formula database

12
PKG Formats
Mass production capacity

Sushi Exclusive

Soy Sauce Boba

Soy × allergen-free Michelin sushi partner

German Hotdog Collab

Double Cucumber Boba

Adventurous cucumber × EU licence

Functional RTD

Energy Drink Boba

Energy drink formula × Vitamin B

Health & Zero-Sugar

Zero Sugar Vegan Boba

100% no added sugar × Vegan cert.

Tlab has solved the zero-sugar membrane-rupture challenge — Zero Added Sugar formula available

Proof-of-Concept (POC) sample development service available

Meeting the World's

Most Stringent Procurement Standards

FSSC 22000

Top-tier food safety management system

ISO 22000

Integrated with HACCP food safety framework

Halal

Full market access: Middle East, Indonesia & beyond

Vegan / Allergen-Free

EU & US health channels — zero allergen guarantee

Operational Strength

24h	66	ERP	RTU
Automated Line	**Export Countries**	**System Control**	**Ready-to-Use**
Human-machine separation	Resilient global supply chain	Accurate,on-time delivery	Zerostore preparation cost

Tachiz maintains a resilient supply chain serving 66 countries & territories worldwide with consistent quality and on-time delivery.

SAVE THIS SLIDE! · 5× Algorithm Boost

2026 RTD & Dessert Topping

R&D Procurement Checklist

5 critical evaluation criteria for topping ingredient suppliers

- ☐ Ready-to-use format? Eliminates store preparation labour & waste — **RTU**
- ☐ Maintains texture & colour at 4°C for 6+ months without hardening? — **Cold**
- ☐ FSSC 22000, ISO 22000 & Halal internationally certified? — **Cert**
- ☐ Supports zero-sugar or functional ingredient R&D customisation? — **Custom**
- ☐ ERP-managed 66-country supply chain with guaranteed delivery? — **Scale**

Tachiz

Tachiz × Tlab Passes ALL Criteria · Zero Compromise

Save for your next procurement review — 1 save = 5× the algorithm weight of a like

TAKE ACTION NOW · BOFU CTA

Ready to Make Your

Next Product Go Viral?

Tachiz Tlab Innovation Lab

Your strongest global food R&D partner
Exporting to 66 countries × 393 custom flavours × World-class certifications

Free Sample Kit — Limited Availability

High-Stability Boba Technology Test Set

Click the link in the FIRST COMMENT below to request your R&D sample form

What's your biggest RTD development challenge —

texture or shelf life? Tell us in the comments!

TACHIZ × TLAB

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